

Smart Study Planner AI

Introduction

The **Smart Study Planner AI** is a Flask-based web application designed to help students organize their study plan efficiently and estimate their probability of completing the syllabus before an exam. The system combines machine learning prediction with automated timetable generation to support structured and goal-oriented learning.

Objectives

- Provide an intelligent system for study schedule planning
- Predict syllabus completion probability using machine learning
- Optimize topic prioritization based on difficulty and time constraints
- Present clear visual analytics for study workload

System Architecture

Technology Stack

Component	Technology
Backend	Flask (Python)
Frontend	HTML, CSS, JavaScript
Machine Learning Model	Logistic Regression (scikit-learn)
Data Processing	Pandas, NumPy
Model Storage	Joblib

Core Functional Modules

Topic Management Module

Users can add and manage study topics by specifying:

- Subject name
- Topic name
- Difficulty level (scale 1–10)

- Estimated study hours required

This information forms the input dataset for scheduling and prediction.

AI-Based Completion Prediction

The system uses a Logistic Regression model trained on synthetic data to estimate the probability of completing all topics before the exam.

Prediction Factors

- Total hours required
- Average topic difficulty
- Days remaining until exam
- Available study hours per day

Output

- Completion probability (0%–100%)
- Color-coded performance indicator

Smart Timetable Generator

The scheduling engine generates an optimized daily study plan based on topic priority.

Priority Calculation Criteria

- Higher difficulty topics
- Topics requiring more time
- Limited time remaining before exam

Scheduling Strategy

- Distributes workload across available days
- Ensures balanced daily study hours
- Produces a structured daily timetable

Visual Analytics Module

The system provides visual insights to improve planning clarity.

Displayed Metrics

- Completion probability indicator
- Total study hours required
- Average difficulty level
- Daily workload bar chart

- Day-wise study schedule

System Workflow

1. User inputs study topics with difficulty and required hours
2. User specifies exam deadline and daily available study time
3. Machine learning model predicts completion probability
4. Scheduler generates optimized study timetable
5. Results are displayed through charts and structured plans

Project File Structure

smart-study-planner-fs/

```
|
├── app.py          # Main Flask application
├── dataset.py      # Synthetic dataset generator
├── model.py        # ML training and prediction
├── scheduler.py    # Timetable generation logic
├── study_data.csv  # Generated training dataset
├── model.pkl       # Trained ML model
├── scaler.pkl      # Feature scaling model
├── requirements.txt # Project dependencies
├──
├── templates/
│   └── index.html  # User interface
├──
├── static/
│   ├── style.css   # UI styling
│   └── script.js   # Frontend logic
```

API Endpoints

Endpoint	Method	Description
/	GET	Renders main interface
/generate-plan	POST	Generates schedule and prediction
/add-topic	POST	Validates topic input
/health	GET	Application health status

Advantages

- Personalized study planning
- Data-driven prediction system
- Automatic priority scheduling
- Simple web-based interface
- Visual decision support

Future Enhancements

- Real user dataset training
- Mobile application integration
- Adaptive learning recommendations
- Progress tracking dashboard
- Cloud deployment support